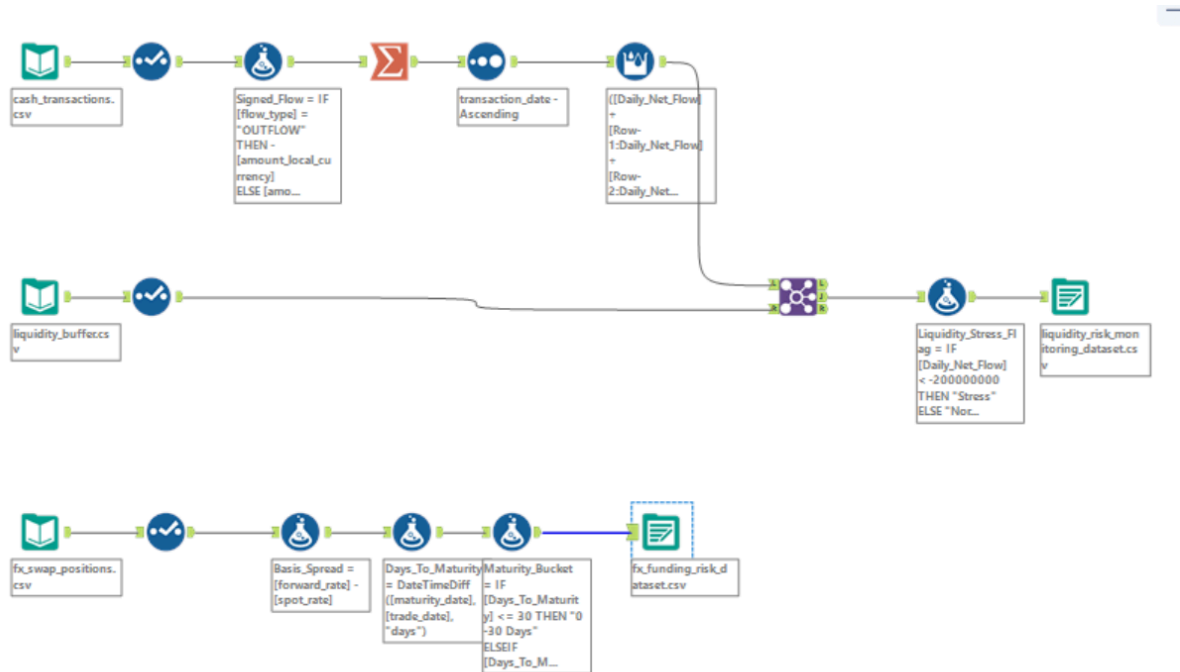




Alteryx Workflow:



Workflow Snapshot

Introduction:

This project is about building an **automated liquidity risk monitoring pipeline** using Alteryx. The goal is to transform raw treasury transaction data into meaningful insights that help the bank monitor cash flow stability, funding risks, and operational settlement performance.

The workflow collects data from multiple sources such as **cash transactions, regulatory liquidity buffers, and FX swap funding positions**. It then cleans and processes the data to calculate important liquidity indicators such as **daily net cash flow, rolling liquidity trends, liquidity stress flags and short-term funding exposure**.

By automating this process, the company can quickly detect unusual liquidity events, understand which business units are driving inflows or outflows and identify short-term funding risks.

The workflow delivers several operational and strategic benefits:

Improved Liquidity Monitoring

The system automatically aggregates and analyzes thousands of treasury transactions, allowing the bank to track daily liquidity movements and detect stress events earlier.

Faster Risk Identification

Rolling liquidity calculations and stress flags highlight abnormal outflows or funding pressures that could affect regulatory liquidity ratios.

Funding Risk Visibility

The FX swap analysis identifies short-term funding concentrations and rollover risks, helping treasury teams manage funding strategies more effectively.

Automated Analytics Pipeline

Instead of manually preparing data, the workflow automatically processes raw data into structured analytical datasets that feed reporting dashboards.

Better Decision Making

Leadership can monitor liquidity health, operational settlement performance and funding exposure through a single integrated analytics system.

Technical Explanation:

The Alteryx workflow implements a multi-source ETL pipeline designed to process treasury and liquidity datasets and generate structured analytical outputs for Tableau dashboards.

The pipeline begins by ingesting raw datasets including cash transactions, liquidity buffer metrics, and FX swap positions. A data cleansing layer standardizes field types, removes unnecessary columns and validates transaction uniqueness to ensure data integrity.

The cash transaction data is transformed to compute signed liquidity flows, converting outflows into negative values for easier aggregation. Using the Summarize tool, the workflow aggregates transaction-level data into daily liquidity metrics by currency and business unit. These daily flows are then sorted and processed through a Multi-Row Formula tool to generate rolling 7-day liquidity averages, enabling trend analysis and volatility monitoring.

Next, the pipeline integrates regulatory liquidity buffer data using a Join tool. This enriches the dataset with LCR and HQLA metrics, allowing the workflow to generate liquidity stress indicators that flag days with abnormal net outflows.

A parallel processing branch analyzes FX swap funding positions. It calculates basis spreads between forward and spot rates, derives days to maturity, and classifies swaps into maturity buckets to evaluate short-term rollover risk and funding concentration.

Finally, the workflow exports two structured datasets:

- Liquidity Monitoring Dataset – containing daily net flows, rolling liquidity trends and regulatory liquidity metrics.
- FX Funding Risk Dataset – containing maturity profiles and funding exposure indicators.

These outputs support downstream analytics and visualization, enabling automated liquidity risk monitoring through Tableau dashboards.

Tools used in this work flow are as follows:

- Input Data
- Select
- Formula
- Summarize
- Sort
- Multi-Row Formula
- Join
- Unique
- Output Data

Usage of Gen AI:

All the data used in this project was artificially generated using GenAI tools (ChatGPT 5.2). A total of 5 datasets were generated for this project.

ChatGPT was also used to fix errors which occurred while developing visuals / calculation fields.

Steps:

1. Perplexity Pro AI was used to find out information on Bank Of America's business using the job description for "Business Data Analyst" and in general operations of Bank of America.
2. This information was then fed to ChatGPT to generate an artificial dataset for project work.
3. The datasets, although being AI generated, have no personal information and are fully GDPR compliant.

The primary purpose of this project was to gain an understanding of the Alteryx Platform and implement a work flow using somewhat similar data that Bank of America deals with. Real world data would be significantly larger and would contain a lot more challenges and have a wider set of tooling required for a workflow.